KRUSHNAKANTA PANDA

Mechatronics Engineering Student | Robotics • Embedded Systems • Industrial Automation
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PROFESSIONAL SUMMARY

Results-driven B.Tech Mechatronics Engineering student at KIIT University (CGPA: 8.1) with hands-on industrial automation exposure from a manufacturing engineering internship at HPCL-Mittal Energy Limited (HMEL), Guru Gobind Singh Refinery — Automated Storage & Retrieval Systems (ASRS) and a Rail Guided Vehicle (RGV) project. Skilled in end-to-end development, from CAD modeling to prototyping and embedded programming, using SolidWorks, AutoCAD, Arduino (C/C++), MATLAB, and 8051 microcontrollers.

WORK EXPERIENCE

HPCL-Mittal Energy Limited (HMEL) — Guru Gobind Singh Refinery (GGSR), Bathinda, Punjab
Manufacturing Engineer Intern — Automated Storage & Retrieval System (ASRS) May 2026 – June 2026

- Completed a 45-day internship (5 May – 20 June 2026) in the Automated Storage & Retrieval System (ASRS) division of HMEL's Guru Gobind Singh Refinery, gaining hands-on exposure to industrial automation and material handling operations.
- Worked on the Rail Guided Vehicle (RGV) project — an automated, rail-mounted material handling vehicle used for high-speed pallet transfer within the refinery's ASRS warehouse.
- Studied and documented core RGV subsystems, including traction drive motors, gearboxes, guide-rail and wheel assemblies, positioning sensors, encoders, and onboard safety interlocks.
- Assisted senior engineers in assembly support, inspection, testing, and preventive maintenance of RGV and ASRS equipment.
- Gained practical exposure to PLC-controlled material flow, sensor-based positioning, and electro-mechanical system integration while following refinery safety and PPE protocols.

KIIT University
Event Organizer & Anchor 2024 – 2025

- Anchored and hosted college fest events, managing stage flow and audience engagement for large crowds.
- Coordinated logistics and team activities as an official event organizer for a dedicated fest segment.
- Executed end-to-end promotional campaigns for events across social media and on-campus channels.

PROJECTS

CNC Plotter | **Arduino, SolidWorks, G-code**

- Designed and fabricated a functional 2-axis CNC plotter capable of drawing vector-based patterns.
- Programmed motion-control logic on Arduino with G-code parsing for precise pen movement.
- Modeled all mechanical components, including frame, carriage, and pen holder, in SolidWorks.

Quadruped Walking Robot (Arduino Dog) | **Arduino, Servo Control**

- Built a four-legged walking robot using servo motors and custom gait logic programmed on Arduino.
- Designed the leg linkage mechanism to achieve stable gait sequences across multiple walking patterns.
- Integrated sensor input for basic obstacle-aware movement and directional control.

Autonomous Mopping Robot | **Arduino, Mechanical Design**

- Developed a ground-cleaning robot capable of navigating a room and mopping dust autonomously.
- Designed the drive mechanism and cloth-dispensing system for consistent surface coverage.
- Programmed path-planning logic on Arduino for efficient area coverage with minimal repetition.

EDUCATION

B.Tech — Mechatronics Engineering 2024 – 2028 (Expected)
Kalinga Institute of Industrial Technology (KIIT), Bhubaneswar | CGPA: 8.1 (up to 4th Semester)

TECHNICAL SKILLS

Design & CAD: SolidWorks, AutoCAD, MATLAB
Programming & Embedded Systems: Arduino (C/C++), Embedded C, 8051 Microcontroller, G-code
Automation & Material Handling: Automated Storage & Retrieval Systems (ASRS), Rail Guided Vehicles (RGV), Industrial Automation, Preventive Maintenance
Manufacturing: CNC Machining, Lathe Operations, Welding, Foundry Techniques (Workshop Trained)
Soft Skills: Leadership, Public Speaking, Event Management, Team Coordination

CERTIFICATIONS & ACHIEVEMENTS

- Certified in 8051 Microcontroller Programming — Coursera.
- Organized and anchored the college fest at KIIT University, managing event execution for a large audience.
- Led project teams in designing and building 3+ working mechatronics prototypes, from CAD concept to final build.

HOBBIES & INTERESTS

Gaming | Exploring New Software & Tools | Hardware Troubleshooting & Repair | Tinkering with Electronics